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Amazon Brazil

Turning Data into Decision

Insight into Customers ,Sellers,Products,
Payments and Fulfillement Performance
Powering Growth in Brazil's E-commerce Market

Customer | Selller | Products | Payments | Fulfillment



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INTRODUCTION

Market Context

Amazon Brazil operates in one of the fastest-growing e-commerce markets in Latin America, generating large volumes of data across the platform.

Data Overview

The platform generates transactional data across customers, sellers, products, orders, payments, and reviews — forming the foundation of this analysis.

Project Focus

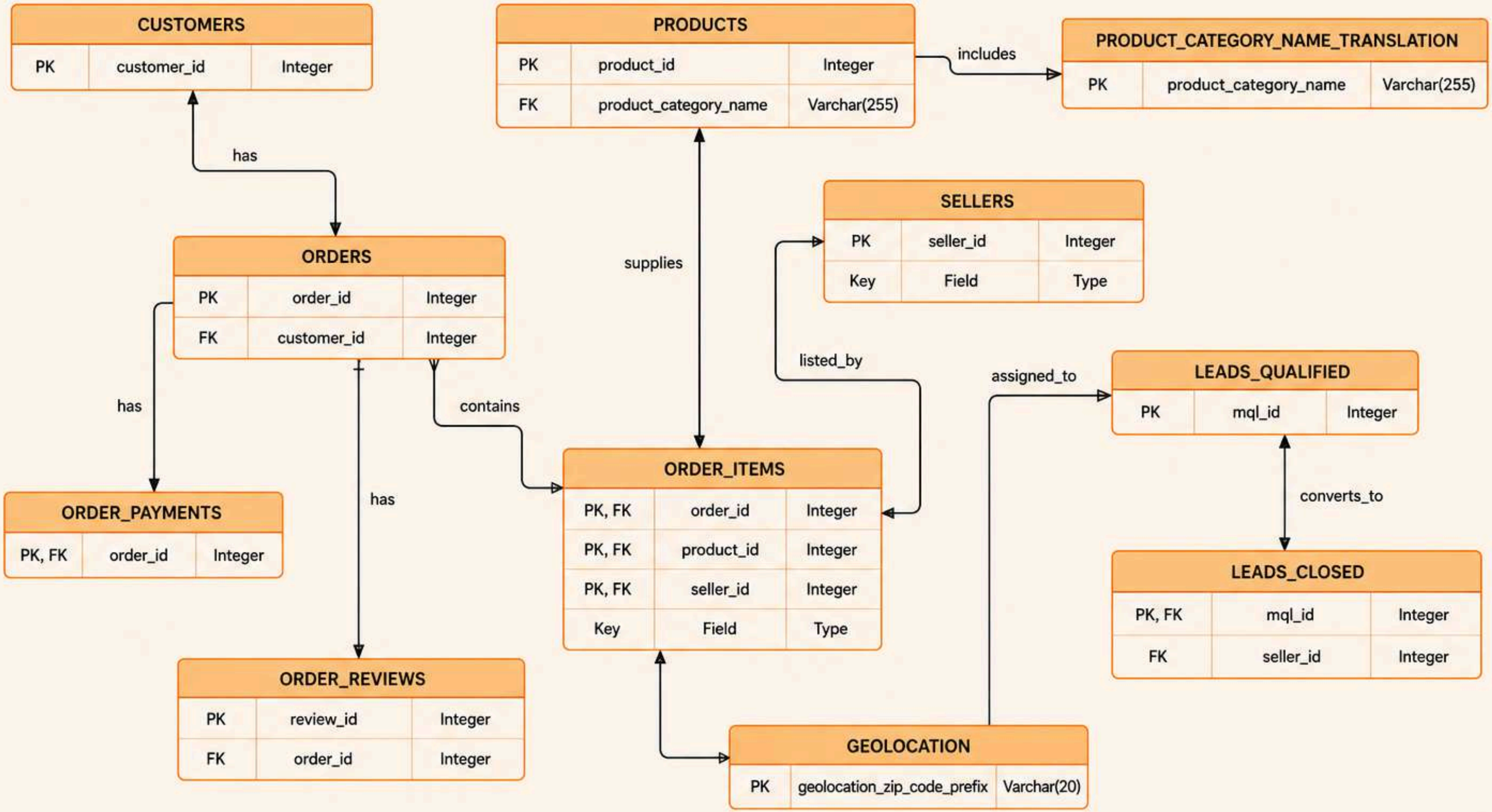
This project focuses on analyzing Amazon Brazil's transactional database using SQL to uncover meaningful business insights across key areas.

Objective

The objective is to evaluate platform performance from multiple perspectives and support data-driven decision-making to drive growth, efficiency, and customer satisfaction.



amazon | DATA SET OVERVIEW & ENTITY RELATIONSHIP



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BUSINESS PROBLEM STATEMENT

Amazon Brazil's marketplace generates large volumes of data across customers, sellers, products, payments, and deliveries. However, extracting meaningful insights to drive business decisions remains a challenge.

Key Business Challenges

Customer Understanding
Limited visibility into customer behavior, purchase patterns, and satisfaction across different regions.

Seller Performance
Difficulty in identifying top-performing sellers and inactive or underperforming sellers.

Product & Category Insights
Lack of clarity on which product categories and items drive the most revenue and which underperform.

Payment Analysis
Need to understand customer payment preferences and identify payment failure trends.

Delivery Efficiency
Delivery times and order fulfillment performance impact customer satisfaction.

Data-Driven Decisions
Absence of consolidated insights makes it difficult to make informed, data-driven business decisions.



Objective

To analyze Amazon Brazil's marketplace data and generate actionable insights that improve customer experience, optimize seller and product performance, enhance delivery efficiency, and drive overall business growth.

Regional Seller Performance



Q1. Find the total number of orders fulfilled by each seller state.

```
SELECT
   seller_state,
   COUNT(DISTINCT oi.order_id) AS total,
   order_status
FROM
   sellers s
      JOIN
   order_items oi ON s.seller_id = oi.seller_id
      JOIN
   orders o ON oi.order_id = o.order_id
GROUP BY seller_state , order_status
HAVING order_status = 'delivered'
ORDER BY total DESC;
```

<> Output

seller_state	total	order_status
SP	68641	delivered
MG	7735	delivered
PR	7512	delivered
RJ	4227	delivered
SC	3603	delivered
RS	1962	delivered
DF	808	delivered
BA	550	delivered
GO	451	delivered
PE	403	delivered
MA	389	delivered

Insight: This analysis identifies the seller states contributing the highest order volumes. The insights help Amazon prioritize logistics investments and fulfillment center expansion in high-demand regions.

Revenue Trend by Product Category



Q2. For each product category, calculate the cumulative revenue generated as orders come in over time.

```
WITH category_revenue AS (
   SELECT
   pct.product_category_name_english AS product_category,
   DATE(o.order_purchase_timestamp) AS order_date,
   SUM(oi.price) AS daily_revenue
FROM
   orders o
      JOIN
   order_items oi ON o.order_id = oi.order_id
      JOIN
   products p ON oi.product_id = p.product_id
      JOIN
   product_category_name_translation pct ON p.product_category_name = pct.product_category_name
GROUP BY pct.product_category_name_english , DATE(o.order_purchase_timestamp)
)
SELECT
   product_category,
   order_date,
   daily_revenue,
   SUM(daily_revenue) OVER (
      PARTITION BY product_category
      ORDER BY order_date
   ) AS cumulative_revenue
FROM category_revenue
ORDER BY
   product_category,
   order_date;
```

<> Output

product_category	order_date	daily_revenu	cumulative_r
agro_industry_and_c...	2017-01-23	43.98	43.98
agro_industry_and_c...	2017-01-31	21.99	65.97
agro_industry_and_c...	2017-02-05	21.99	87.96
agro_industry_and_c...	2017-02-08	21.99	109.949999..
agro_industry_and_c...	2017-02-12	92.9	202.85
agro_industry_and_c...	2017-02-13	21.99	224.84
agro_industry_and_c...	2017-02-16	21.99	246.83
agro_industry_and_c...	2017-02-18	21.99	268.82

Insight: Cumulative revenue analysis shows how sales build over time for each product category. It helps Amazon Brazil identify fast-growing categories, optimize promotional campaigns, improve inventory planning, and make data-driven decisions to maximize revenue.

Payment Method Performance



Q3. Which payment method do customers use the most, and what is the average order value for each payment type.

```
SELECT
    payment_type,
    COUNT(payment_type) AS total,
    AVG(payment_value) AS average
FROM
    order_payments
GROUP BY payment_type
ORDER BY total DESC;
```

<> Output

payment_type	total	average
credit_card	76795	163.31902063935578
boleto	19784	145.0344354023441
voucher	5775	65.70335411255424
debit_card	1529	142.57017004578162
not_defined	3	0

Insight: This analysis identifies the most popular payment method and compares average order values across payment types. The findings help Amazon optimize payment partnerships, reduce transaction costs, and target high-value customer segments.

High-Value Customer Analysis



Q4. Find the customer who has spent the most money across all their orders.

```
SELECT
   customer_unique_id, SUM(op.payment_value) AS
total_spent
FROM
   customers c
      JOIN
   orders o ON c.customer_id = o.customer_id
      JOIN
   order_payments op ON o.order_id = op.order_id
GROUP BY customer_unique_id
ORDER BY total_spent DESC
LIMIT 1;
```

<> Output

customer_unique_id	total_spent
0a0a92112bd4c708ca5fde585afaa872	13664.08

Insight: This analysis identifies the highest-spending customer on the platform based on total purchase value. The insights help Amazon launch its premium loyalty program and better understand the behavior of its most valuable customers.

Customer Satisfaction by Category



Q5. Find the average review score for each product category.

```
SELECT
    pct.product_category_name_english,
    AVG(review_score) AS average_review
FROM
    products p
        JOIN
    order_items oi ON p.product_id = oi.product_id
        JOIN
    order_reviews orw ON oi.order_id = orw.order_id
        JOIN
    product_category_name_translation pct
        ON p.product_category_name =
pct.product_category_name
GROUP BY product_category_name_english
ORDER BY average_review DESC;
```

<> Output

product_category_name_english	average_review
cds_dvds_musicals	4.6429
fashion_childrens_clothes	4.5000
books_general_interest	4.4463
costruction_tools_tools	4.4444
flowers	4.4194
books_imported	4.4000
books_technical	4.3684

Insight: These insights help prioritize categories that need immediate intervention and support data-driven decisions to improve overall customer experience.

Customer Purchase Distribution



Q6. Find the total number of orders placed by each customer, broken down by the state they live in.

```
SELECT
   c.customer_unique_id,
   c.customer_state,
   COUNT(o.order_id) AS total_order
FROM
   customers c
      JOIN
   orders o ON c.customer_id = o.customer_id
GROUP BY customer_state , customer_unique_id
ORDER BY total_order DESC;
```

<> Output

customer_unique_id	customer_state	total_order
8d50f5eadf50201ccdcedfb9e2ac8455	SP	17
3e43e6105506432c953e165fb2acf44c	SP	9
6469f99c1f9dfae7733b25662e7f1782	SP	7
1b6c7548a2a1f9037c1fd3ddfed95f33	MG	7

Insight: The analysis identified variations in customer order frequency across Brazilian states, highlighting regions with the highest and lowest levels of customer engagement. These insights help optimize regional marketing and loyalty strategies.

Dormant Seller Analysis



Q7. Identify sellers who registered on the platform but have never fulfilled a single order.

```
SELECT
    s.seller_id, s.seller_city
FROM
    sellers s
        LEFT JOIN
    order_items oi ON s.seller_id = oi.seller_id
WHERE
    oi.seller_id IS NULL;
```

<> Output

Action	Message
SELECT customer_unique_id, SUM(op.payment_value) AS total_spent FROM customers c JOIN ...	1 row(s) returned
SELECT c.customer_unique_id, c.customer_state, COUNT(o.order_id) AS total_order FROM cus...	1000 row(s) returned
SELECT s.seller_id, s.seller_city FROM sellers s LEFT JOIN order_items oi ON s.seller_id = oi.s...	0 row(s) returned

Insight: The analysis identified inactive sellers who have never fulfilled an order despite being registered on the platform. These accounts represent potential opportunities for seller activation or platform cleanup.

Revenue Leaders by Category



Q8. Find the top 5 product categories by total revenue.

```
SELECT
   product_category_name_english,
   ROUND(SUM(op.payment_value), 2) AS revenue
FROM
   product_category_name_translation pct
      JOIN
   products p ON pct.product_category_name =
p.product_category_name
      JOIN
   order_items oi ON oi.product_id = p.product_id
      JOIN
   order_payments op ON op.order_id = oi.order_id
GROUP BY product_category_name_english
ORDER BY revenue DESC
LIMIT 5;
```

<> Output

product_category_name_english	revenue
bed_bath_table	1712553.67
health_beauty	1657373.12
computers_accessories	1585330.45
furniture_decor	1430176.39
watches_gifts	1429216.68

Insight: These insights help allocate promotional budgets effectively, optimize inventory levels, and prioritize high-performing categories for homepage placement during major sales events .

Order Delivery Performance



Q9. Find the median delivery time (in days) between order placement and actual delivery.

```
WITH delivery_days AS (
   SELECT
   DATEDIFF(order_delivered_customer_date,
         order_purchase_timestamp) AS delivery_time
FROM
   orders
WHERE
   order_delivered_customer_date IS NOT NULL
)
SELECT AVG(delivery_time) AS median_delivery_time
FROM (
   SELECT
      delivery_time,
      ROW_NUMBER() OVER (ORDER BY delivery_time) AS rn,
      COUNT(*) OVER () AS total_rows
   FROM delivery_days
) t
WHERE rn IN (
   FLOOR((total_rows + 1) / 2),
   FLOOR((total_rows + 2) / 2)
);
```

<> Output

median_delivery_time
10.0000

Insight: The median delivery time reflects the typical time customers wait for their orders, providing a reliable measure of delivery performance. By focusing on this, Amazon Brazil can identify logistics inefficiencies, optimize fulfillment operations, and improve the overall customer experience.

Product Catalog Optimization



Q10. Find all products that have never been ordered.

```
SELECT
   p.product_id, p.product_category_name
FROM
   products p
      LEFT JOIN
   order_items oi ON p.product_id = oi.product_id
WHERE
   oi.product_id IS NULL;
```

<> Output

Action	Message
SELECT product_category_name_english, ROUND(SUM(oi.price), 2) AS revenue FROM product_c...	5 row(s) returned
SELECT c.customer_unique_id, c.customer_state, COUNT(o.order_id) AS total_order FROM cus...	1000 row(s) returned
SELECT p.product_id, p.product_category_name FROM products p LEFT JOIN order_items oi...	0 row(s) returned

Insight: These insights help optimize the product catalog by removing or improving underperforming listings, reducing catalog clutter, improving customer browsing experience, and focusing attention on products with higher sales potential.

Seller Performance Benchmark



Q11. Find sellers who have fulfilled more orders than the average seller on the platform.

```
WITH highest AS (
SELECT
   s.seller_id, COUNT(DISTINCT oi.order_id) AS total_count
FROM
   sellers s
      JOIN
   order_items oi ON s.seller_id = oi.seller_id
GROUP BY seller_id
ORDER BY total_count DESC)

SELECT *
FROM
   highest
WHERE   total_count > (SELECT AVG(total_count)
      FROM highest);
```

<> Output

seller_id	total_co
6560211a19b47992c...	1854
4a3ca9315b744ce9f8...	1806
cc419e0650a3c5ba77...	1706
1f50f920176fa81dab9...	1404
da8622b14eb17ae283...	1314
955fee9216a65b617a...	1287
7a67c85e85bb2ce858...	1160

Insight: These high-performing sellers are strong candidates for Amazon Brazil's "Top Seller" badge program. The findings also establish a clear performance benchmark that can be used to motivate and evaluate other sellers on the platform.

Regional Customer Experience



Q12. Find which Brazilian states have the highest average customer review score for orders delivered there.

```
SELECT
   c.customer_state,
   ROUND(AVG(r.review_score), 2) AS avg_review_score
FROM
   customers c
      JOIN
   orders o ON c.customer_id = o.customer_id
      JOIN
   order_reviews r ON o.order_id = r.order_id
GROUP BY c.customer_state
ORDER BY avg_review_score DESC;
```

<> Output

customer_state	avg_review_score
AP	4.19
PR	4.18
AM	4.18
SP	4.17
MG	4.14
RS	4.13
MS	4.12
RN	4.11

Insight: These insights help Amazon identify regions where logistics operations are performing well and areas where delivery partners may require improvement to enhance customer satisfaction and maintain service quality.

Review Participation Analysis



Q13. Identify customers who have placed orders but never left a review.

```
SELECT DISTINCT
   c.customer_unique_id, review_score
FROM
   customers c
      JOIN
   orders o ON c.customer_id = o.customer_id
      LEFT JOIN
   order_reviews r ON o.order_id = r.order_id
WHERE
   r.review_id IS NULL;
```

<> Output

customer_unique_id	review_score
618ab23cb329ac11c84bc092ddb4b...	NULL
2990f218665832600f71806507030...	NULL
d780d06a532da7d8d0435cd348e59...	NULL
a5383a6d51465659e9153a2a6045e...	NULL
a7b4c690609bea24107020ae85cc5...	NULL
479cc370a1fdb7d10dfe46016182a1...	NULL
2e9cdecaa137e8ba2084bbde28c85...	NULL
ea870f4fdfd85ac98ab775b76efe3143	NULL
367f4686d7112d69feed92b02a1775...	NULL
528b011eb7fab3d59c336cc7248eed...	NULL
66afa2376b0e4599e420694f53eae80a	NULL

Insight: These insights enable targeted review reminder campaigns, helping Amazon increase product reviews, improve product discoverability, strengthen buyer trust, and enhance the overall shopping experience.

Seasonal Order Trend Analysis



Q14. Find the month with the highest number of orders placed across the entire platform.

```
SELECT
   YEAR(order_purchase_timestamp) AS year_,
   MONTH(order_purchase_timestamp) AS month_,
   COUNT(order_id) AS total_orders
FROM
   orders
GROUP BY year_ , month_
ORDER BY total_orders DESC
LIMIT 1;
```

<> Output

year_	month_	total_orders
2017	11	7544

Insight: These insights help forecast future demand, optimize warehouse capacity, plan staffing requirements, and schedule marketing campaigns ahead of peak shopping periods to ensure smooth operations and maximize sales opportunities.

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Recommendation

Optimize Marketing Strategies

Target high-value customer segments with personalized campaigns to increase conversion and retention.

Empower Top Performing Sellers

Support high-performing sellers with incentives, training, and better visibility opportunities.

Focus on High Demand Categories

Increase inventory availability and optimize pricing for top-performing product categories.

Improve Payment Success Rate

Analyze payment failure patterns and enhance payment experience to reduce drop-offs.

Strengthen Fulfillment Operations

Reduce delivery time and improve order tracking to enhance customer satisfaction.

Leverage Reviews for Growth

Use customer feedback to improve products, seller performance, and overall experience.



CONCLUSION

Actionable Insights

The analysis transformed Amazon Brazil's transactional data into actionable insights across key business areas.

Business Impact

Insights across sales, customers, sellers, products, logistics, and reviews help optimize performance and drive growth.

Operational Efficiency

Data-driven findings support better fulfillment operations, reduce delivery time, and improve customer satisfaction.

Stronger Performance

The analysis identifies top-performing sellers, high-revenue categories, and key improvement areas.

Future Growth

These insights empower data-driven decision-making, enabling Amazon Brazil to scale efficiently and stay ahead in the competitive e-commerce market.



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THANK YOU !

Thank for your time and attention.

I hope this analysis provides valuable insights and support data-driven decision-making for Amazon Brazil.